

EXPERIMENT 4

HILL CIPHER

AIM

To implement the Hill Cipher substitution technique using C.

ALGORITHM

1. Read the plaintext from the user.
2. Read the 3×3 key matrix.
3. Convert the plaintext into numerical values ($A=0, B=1, \dots, Z=25$).
4. Multiply the plaintext vector with the key matrix modulo 26.
5. Convert the result back to characters to obtain the cipher text.
6. Display the encrypted text.

CODE

```
#include <stdio.h>

int main() {
    int key[2][2], text[2], cipher[2];

    printf("Enter 2x2 Key Matrix:\n");
    scanf("%d%d%d%d",&key[0][0],&key[0][1],&key[1][0],&key[1][1]);

    char msg[3];
    printf("Enter 2-letter Plain Text: ");
    scanf("%s",msg);

    text[0]=msg[0]-'A';
    text[1]=msg[1]-'A';

    cipher[0]=(key[0][0]*text[0]+key[0][1]*text[1])%26;
    cipher[1]=(key[1][0]*text[0]+key[1][1]*text[1])%26;

    printf("Encrypted Text: %c%c\n",cipher[0]+'A',cipher[1]+'A');

    return 0;
}
```

OUTPUT

```
Output Clear  
Enter 2x2 Key Matrix:  
2 5  
4 9  
Enter 2-letter Plain Text: hi  
Encrypted Text: SW
```

RESULT

Thus the Hill Cipher substitution technique was implemented successfully using C.